

On Optimization of Mixed-Radix FFT: A Signal Processing Approach

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Abstract—In order to achieve higher transmission rates and system capacities, fifth generation (5G) systems as well as 802.11 ad/ay systems consider higher frequency bands (24GHz-70GHz), the so-called millimeter wave frequencies. These systems rely heavily on Orthogonal Frequency Division Multiplexing (OFDM) approach in order to realize the transmitted signal. The Fast Fourier Transform (FFT) and inverse FFT (IFFT) blocks are vital signal processing components in synthesizing the OFDM signal. Towards the objective of designing a low-complexity, small area, and low-power consumption FFT blocks, this paper proposes an extension of the algorithmic level approach, known as COordinate Rotation Digital Computer (CORDIC)-Friendly FFT in [1], to a more practical higher order FFTs and mixed-radix FFTs. Moreover, it introduces the first quantization error analysis of a generalized mixed-radix FFT. The error analysis is used to find the optimal modified twiddle factors that reduce the FFT hardware complexity. Additionally, this paper shows that any type of CORDIC considered in the proposed framework outperforms the conventional FFT in terms of Signal-to-Quantization-Noise Ratio (SQNR), for a given silicon area budget. The proposed framework offers up to 50 dB SQNR gain compared to the conventional FFT for different FFT sizes when CORDIC rotators are employed. If Single-path Delay Feedback (SDF) pipeline architecture is used, the achieved SQNR gain is obtained at no additional hardware cost.

Index Terms—CORDIC, FFT, Mixed-Radix, SDF, OFDM.

I. INTRODUCTION

Fast Fourier Transform (FFT) is a basic building block in many applications. Most of today's wireless communications standards are based on OFDM (Orthogonal Frequency Division Multiplexing) and OFDMA (Orthogonal Frequency Division Multiple Access). Both technologies are based on making an inverse FFT (IFFT) at the transmitter side, and FFT at the receiver side. Fifth Generation (5G), Long Term Evolution (LTE), WiFi, WiGig, Digital Video Broadcasting standards (DVB-T, DVB-T2, DVB-H) and many other wireless standards are all based on OFDM or OFDMA. FFT is also employed in all applications that need spectral analysis such as noise filtration, remote sensing, image enhancement and compression, and others. In many of the previously mentioned applications, power consumption and speed of the FFT is of major consideration. For example, in wireless communication, reducing power consumption of all receiver blocks, including

the FFT, helps in extending the battery life of the devices. This is especially important for devices that need extended battery life such as Body Area Network (BAN) sensors that rely on energy harvesting. Hence, reducing the complexity of the FFT implementation is translated to working in a better power-speed-area performance point.

In FFT, complex multipliers are necessary to synthesize the twiddle factors rotation operations in each FFT stage. However, the rotations can be synthesized using COordinate Rotation Digital Computer (CORDIC) algorithm to reduce the complexity of FFT implementation. The CORDIC algorithms suggests dividing the target rotation into series of micro rotations. The micro rotations are performed using shift and add/sub operations. Hence, a multiplication-free FFT implementation can be achieved, which reduces the FFT hardware complexity. Since the invention of the first CORDIC algorithm [2], many other CORDIC types have emerged over years [3]–[5]. In the conventional FFT architecture, equivalent CORDIC gains is required for all the synthesized rotations. This constraint is enforced in order to avoid gain compensation after each FFT stage. Equal CORDIC gains for different rotations can be obtained by the original CORDIC algorithm in [2] and Mixed Scaling Rotation (MSR) CORDIC in [5]. However Modified Vector Rotation (MVR) CORDIC [3] and Extended Elementary Angle Set (EEAS) CORDIC [4] algorithms introduce unequal CORDIC gains for different target rotations. Therefore, the original CORDIC algorithm and MSR CORDIC are the best choices for authors implementing CORDIC-based FFT [6], [7]. In [1], authors proposed a CORDIC-friendly FFT architecture. The modified FFT architecture supports using MVR and EEAS CORDIC algorithms without gain compensation after each FFT stage.

The first contribution of this paper is an extension of radix-2 CORDIC-Friendly FFT in [1] to mixed-radix FFT which consists of different radices values at different FFT stages. The second contribution is a novel quantization error analysis for radix- r CORDIC-friendly butterfly. The error analysis helps in finding the optimal CORDIC rotations, which achieve the highest Signal-to-Quantization-Noise Ratio (SQNR), without increasing the hardware resources. To the best of our knowledge, this is the first proposed mixed radix FFT that allows the usage of the efficient MVR and EEAS CORDIC types.

This work was supported by the Egyptian Information Technology Industry Development Agency (ITIDA) under the ITAC Program CFP#84.

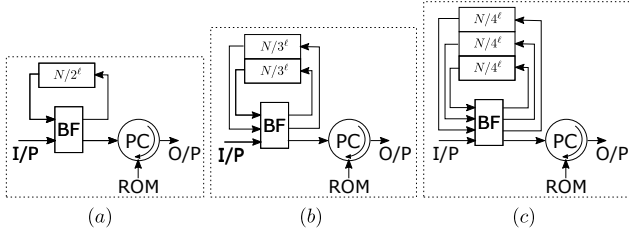


Fig. 1: Single-path delay feedback processing elements for (a) radix-2, (b) radix-3 and (c) radix-4.

The third contribution is identifying the optimal CORDIC type attaining the highest SQNR for a silicon area budget.

The remainder of this paper is organized as follows. Section II reviews the serial pipeline FFT architecture. Section III illustrates briefly the motivation of the algorithm devised in this paper. Section IV and V illustrate the system model and the error analysis of the proposed model. Section VI shows the enhancements offered by the proposed radix- r optimization for different FFT sizes in the SQNR and hardware complexity. Section VII draws the conclusion.

II. RADIX- r FFT HARDWARE ARCHITECTURES

Different applications impose wide-range of specifications on FFT implementation. In high throughput applications, the pipeline-based architectures are the most suitable [8]. Pipeline-based architectures are divided into two types, parallel pipeline and serial pipeline architectures [9], [10]. In serial pipeline architecture, the hardware cost is minimized as well as the I/O requirements. However, parallel-pipeline architecture achieves higher throughput and lower latency at the cost of substantial increase in the number of memory elements, computational units and I/O requirements. In communication systems, the massive throughput offered by parallel-pipeline architecture is not needed. On the other hand, the silicon area and power consumption are required to be minimized. Therefore when it is related to communication systems, serial pipeline is a suitable design choice [8], [11]–[13].

In the literature, there are two types of serial pipeline architectures: Single-path Delay Feedback (SDF) and Single-path Delay Commutator (SDC). SDF architecture is more efficient, by 50%, in terms of memory usage compared to SDC [8]. Therefore, SDF architecture is adopted in this investigation. Fig. 1 shows the processing elements of the SDF architecture for radix-2, 3 and 4, denoted as R2SDF, R3SDF and R4SDF respectively. In Fig. 1, BF is the butterfly block, PC is pipeline CORDIC block, N is the FFT size, and $\ell \in \{1, \dots, \log_r(N)\}$ is the FFT stage index. The overall pipeline FFT processor is composed of several series processing elements. The number of processing elements is equal to the number of FFT stages. Fig. 2 (a) and (b) illustrates Decimation in Time (DIT) 12-point conventional mixed radix FFT Signal Flow Graph (SFG) and its corresponding SDF architecture, where $W_N^n = e^{-j2\pi n/N}$ are the twiddle factors. In mixed radix FFT, different radix decomposition is employed in each FFT stage. For example in Fig. 2 (a), the first stage employs radix-3 decomposition while radix-2 is employed in the remaining

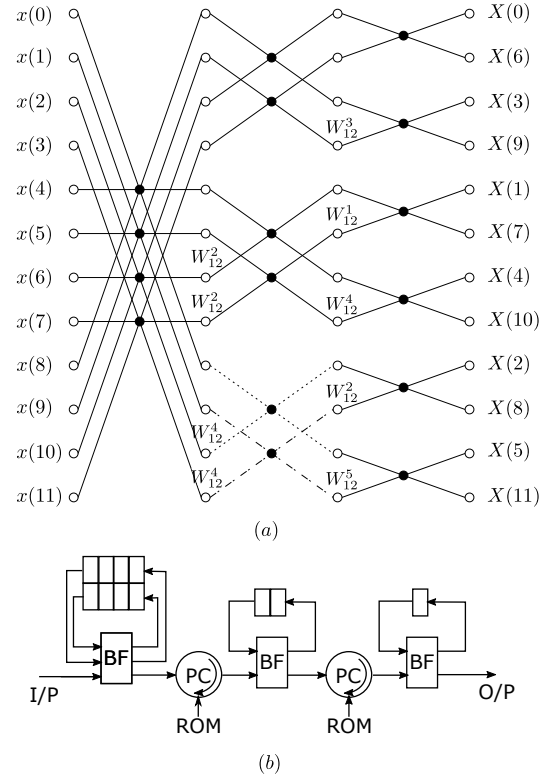


Fig. 2: (a) 12-point DIT mixed radix conventional FFT. (b) SDF architecture for 12-point FFT size.

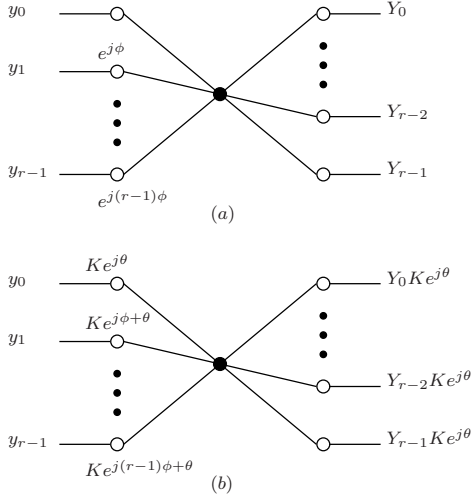
stages. The CORDIC rotations can be pre-calculated from the FFT SFG and stored in Read Only Memory (ROM) to be used in operation.

III. MOTIVATION

In SDF architectures explained in Section II, the outputs of all butterflies belonging to the same FFT stage are rotated using the same CORDIC, as shown in Fig. 1 and 2. In the conventional FFT architecture, PC is not fully utilized because there exists some twiddle-factor-free butterfly nodes. In [1], authors proposed a modification to FFT SFG in order to exploit this opportunity. They introduce twiddle factors at twiddle-factor-free nodes in order to increase the SQNR. Even though the total number of twiddle factors is increased in the SFG, the hardware cost to synthesize the CORDIC rotations is the same as the conventional FFT. On the other hand, the SQNR of the system is substantially increased.

This work generalizes the idea introduced in [1] to mixed-radix FFT. Radix- r and mixed-radix FFTs are of great significance, specially to communication systems. For example radix-4 is commonly used instead of radix-2 because it requires less number of twiddle factors multiplications [8], [11]. Furthermore, radix-3 and radix-5 are used in LTE systems to support non-power-of-two transform lengths [14].

The basic idea of the proposed optimization scheme is depicted in Fig. 3. The conventional twiddle factors are $(e^{j\phi}, e^{j2\phi}, \dots, e^{j(r-1)\phi})$, where ϕ is a fixed angular spacing between twiddle factors which is defined for each butterfly.


 Fig. 3: Radix- r CORDIC-Friendly FFT degree of freedom.

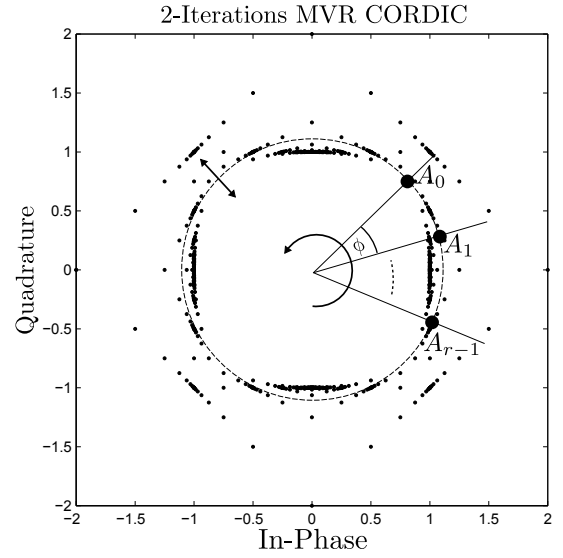
The butterfly operation is only sensitive to the phase difference between the butterfly input operands [1]. Therefore, any complex gain multiplied at the input nodes will propagate to the output nodes as depicted in Fig. 3 (b). Hence, instead of being bound to the original twiddle factors ($e^{j\phi}, e^{j2\phi}, \dots, e^{j(r-1)\phi}$), the modified architecture suggests synthesizing new rotations having equal complex gains $Ke^{j\theta}$ at all input nodes. The degree of freedom introduced by the modified FFT architecture allows achieving enhancements in any of these axes: area, power, speed or flexibility. For instance, multiplication with some numbers is easier than the others; i.e., multiplication by 2^M , where M is integer, is only a shift operation. By choosing the suitable objective function in the optimization problem, better performance than the conventional mixed-radix FFT is expected.

IV. RADIX-R OPTIMIZATION SCHEME SYSTEM MODEL

The outputs of radix- r butterfly are governed by the following equation,

$$Y_\mu = \sum_{\nu=0}^{r-1} W_r^{\nu\mu} y_\nu e^{j\nu\phi} \quad (1)$$

where $e^{j\nu\phi}$ are the twiddle factors attached with the butterfly nodes and $W_r^{\nu\mu} = e^{-j2\pi\nu\mu/r}$. Input index $\nu \in \{0, 1, \dots, r-1\}$ and output index $\mu \in \{0, 1, \dots, r-1\}$. The system model of radix- r butterfly operation is depicted in Fig 3, where y_ν and Y_μ represent the butterfly inputs and outputs, respectively. The key idea of the proposed algorithm is in the process of synthesizing the new CORDIC rotations at the input butterfly nodes. Fig. 4 illustrates the optimization methodology of the proposed FFT algorithm. The small dots in the figure represent the reachable CORDIC rotations, where x-axis and y-axis represent the in-phase and quadrature components of the CORDIC rotations. Without loss of generality, the reachable CORDIC rotations of 2-iteration MVR CORDIC [3] is used for illustration. The Modified Twiddle Factors (MTFs) are represented by the large points located on the intersection between the arms and the circle. The MTFs are denoted as


 Fig. 4: Optimization methodology of the radix- r CORDIC-Friendly FFT

$A_0, A_1, \dots, A_{r-1}$ which are equivalent to $Ke^{j\theta}, Ke^{j\phi+\theta}, \dots, Ke^{j(r-1)\phi+\theta}$ in Fig. 3 (b). The MTFs must have the same complex gain, i.e., must exist on the same circle. Moreover, to guarantee correct butterfly operation, the phase differences between the MTFs are the same as the phase differences between the original twiddle factors.

In the conventional radix- r FFT, the twiddle factor A_0 is equal to one, and the optimization methodology searches for the optimal feasible CORDIC rotations for the remaining twiddle factors, i.e. A_1 to A_{r-1} . Hence, the remaining twiddle factors are enforced to be located on the unit circle and in fixed pre-determined angular positions. However, in the proposed radix- r CORDIC-Friendly FFT, the optimization methodology searches for r feasible CORDIC rotations that are ϕ radians apart from their closest neighbors and located on the same circle as well. The optimal points that satisfy these conditions, are the ones which minimize the quantization error, as will be shown in Section V.

A simple way to summarize the proposed radix- r optimization methodology is given below.

- Changing the radius of the circle along with rotating the twiddle factor arms.
- Finding the nearest feasible CORDIC points to the intersection point of the circle with the arms, i.e., the MTFs.

In order to keep track with the notations defined in [1], the complex gain multiplied by all the butterfly's outputs $Ke^{j\theta}$ is called Propagating Twiddle Factor (PTF). The MTFs are physically synthesized at each FFT stage using CORDIC. On the other hand, PTFs generated from each butterfly propagate to the last stage, merge with the other PTFs from different butterflies after the last FFT stage. Then these scale factors are compensated using one compensation stage at the end.

Moving the PTFs to the last FFT stage is possible because of the special structure of FFT SFG. To illustrate the propagation of the PTFs to the last FFT stage, a 12-point mixed radix

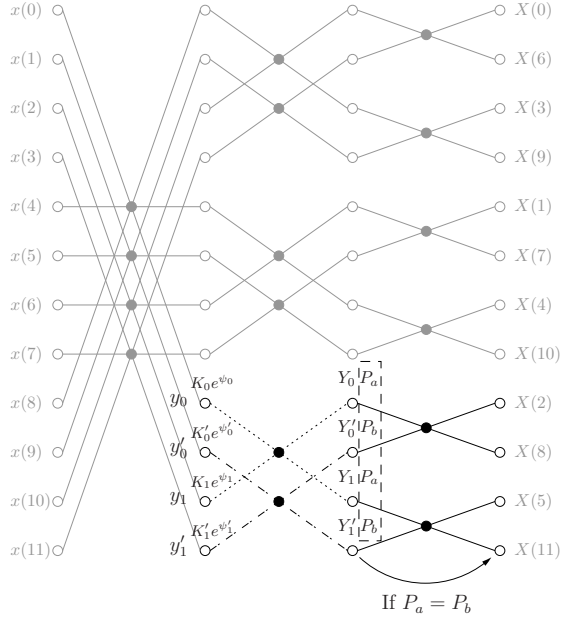


Fig. 5: Transferring the PTFs to the last FFT stage illustration.

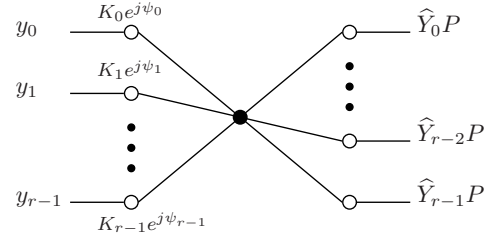
FFT is considered as an example for simplicity. The SFG is depicted in Fig. 2 (a). The two dashed butterflies belong to the same butterfly group because similar original twiddle factors are attached to their inputs as shown in Fig 2 (a). This discussion will focus on these butterflies, the other butterflies will be omitted for simplicity, as depicted in Fig. 5. The MTFs for the dashed butterflies are $K_0 e^{j\psi_0}$, $K'_0 e^{j\psi'_0}$, $K_1 e^{j\psi_1}$ and $K'_1 e^{j\psi'_1}$. The PTFs are P_a and P_b . If the two butterflies have been optimized differently, their MTFs will be different and hence their PTFs will be different as well. Then, it is impossible to transfer these PTFs to the last stage. However, since they belong to the same butterfly group, they will be optimized similarly. As a result, their PTFs will be equivalent. Hence the PTFs P_a and P_b can propagate through the butterflies in the subsequent FFT stages as depicted in Fig. 5. Similarly, all the PTFs from the other butterfly groups can be transferred to the last stage, merged together and then compensated using one compensation stage.

V. CORDIC-FRIENDLY RADIX- r BUTTERFLY ERROR ANALYSIS

Fig. 6 shows the CORDIC Friendly radix- r butterfly operation where $y_0, y_1, \dots, y_{r-1}$ are the input, $K_0 e^{j\psi_0}, K_1 e^{j\psi_1}, \dots, K_{r-1} e^{j\psi_{r-1}}$ are the MTFs and P is the PTF. The output of the butterfly operation is denoted as $\hat{Y}_0, \hat{Y}_1, \dots, \hat{Y}_{r-1}$. Similar to the conventional radix- r butterfly, the outputs of the CORDIC-friendly radix- r butterfly is governed by the following equation,

$$\hat{Y}_\mu P = \sum_{\nu=0}^{r-1} W_r^{\nu\mu} y_\nu K_\nu e^{j\psi_\nu}. \quad (2)$$

The reference radix- r butterfly model is depicted in Fig. 3 (a). Error is defined as the difference between the proposed and


 Fig. 6: Radix- r CORDIC-Friendly butterfly operation

reference model as follows,

$$E_\mu = Y_\mu - \hat{Y}_\mu \quad (3)$$

$$E_\mu = \sum_{\nu=0}^{r-1} \overbrace{W_r^{\nu\mu} y_\nu e^{j\nu\phi}}^{\text{reference model}} - \sum_{\nu=0}^{r-1} W_r^{\nu\mu} y_\nu \frac{K_\nu e^{j\psi_\nu}}{P} \quad (4)$$

$$E_\mu = \sum_{\nu=0}^{r-1} W_r^{\nu\mu} y_\nu \left(e^{j\nu\phi} - \frac{K_\nu e^{j\psi_\nu}}{P} \right). \quad (5)$$

Let $K_\nu = e^{\alpha_\nu}$ and $P = e^\beta$ where $\alpha \in \mathbb{R}$ and $\beta \in \mathbb{C}$, hence

$$E_\mu = \sum_{\nu=0}^{r-1} W_r^{\nu\mu} y_\nu \left(e^{j\nu\phi} - \frac{e^{\alpha_\nu + j\psi_\nu}}{e^\beta} \right) \quad (6)$$

$$E_\mu = \sum_{\nu=0}^{r-1} W_r^{\nu\mu} y_\nu e^{j\nu\phi} \left(1 - \frac{e^{\alpha_\nu + j\psi_\nu}}{e^{\beta + j\nu\phi}} \right) \quad (7)$$

$$E_\mu = \sum_{\nu=0}^{r-1} W_r^{\nu\mu} y_\nu e^{j\nu\phi} \left(1 - \frac{e^{\Omega_\nu}}{e^\beta} \right) \quad (8)$$

$$E_\mu = \sum_{\nu=0}^{r-1} W_r^{\nu\mu} y_\nu e^{j\nu\phi} (1 - e^{\Omega'_\nu}) \quad (9)$$

where $\Omega_\nu = \alpha_\nu + j(\psi_\nu - \nu\phi)$ and $\Omega'_\nu = \Omega_\nu - \beta$. Therefore the error variance can be computed according to

$$\sigma_{E_\mu}^2 = \sigma_X^2 \sum_{\nu=0}^{r-1} |1 - e^{\Omega'_\nu}|^2. \quad (10)$$

Hypothetically, if Ω'_ν was continuous, the optimal solution that minimize the error variance should be at $\Omega'_\nu = 0$, and hence at $\Omega_0 = \Omega_1 = \dots = \Omega_{r-1} = \beta$. However, Ω'_ν is function in $\alpha_\nu + j\psi_\nu$ which belongs to the discrete set of reachable CORDIC rotations denoted as $\mathcal{M}$. For a given Ω_ν , choosing the best β that minimize the objective function can be obtained analytically as shown in the next steps.

The error variance expression in (10) can be written as

$$\sigma_{E_\mu}^2 = \sigma_X^2 \sum_{\nu=0}^{r-1} \left| 1 - \frac{e^{\Omega_\nu}}{e^\beta} \right|^2 \quad (11)$$

$$\sigma_{E_\mu}^2 = \sigma_X^2 \sum_{\nu=0}^{r-1} \left(1 - \frac{e^{\Omega_\nu}}{e^\beta} \right) \left(1 - \frac{(e^{\Omega_\nu})^*}{(e^\beta)^*} \right). \quad (12)$$

To find the value of e^β which minimize the error variance, differentiation with respect to e^β is required. In [15], the author

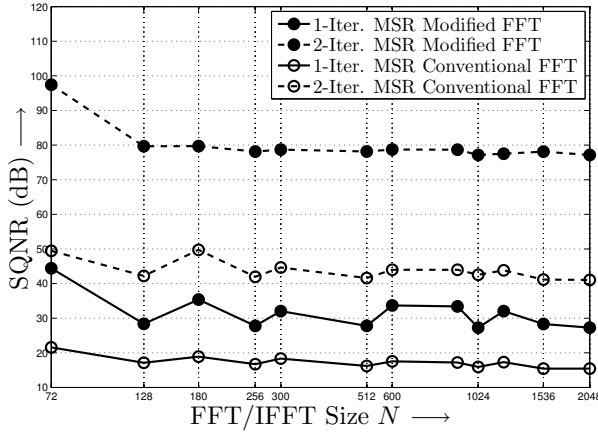


Fig. 7: Comparison between mixed-radix conventional and modified FFT architecture employing MSR CORDIC for different FFT/IFFT sizes.

suggests decomposing the function $\sigma_{E_\mu}^2(e^\beta)$ to $\sigma_{E_\mu}^2(e^\beta, (e^\beta)^*)$ as shown in (12). Afterwards, differentiation with respect to $(e^\beta)^*$ is performed as follows,

$$\frac{d\sigma_{E_\mu}^2}{d(e^\beta)^*} = 0 \quad (13)$$

$$\sum_{\nu=0}^{r-1} (e^{\Omega_\nu})^* (e^\beta - e^{\Omega_\nu}) = 0.$$

Finally e^β which minimize the error variance can be computed according to

$$e^\beta = \frac{\sum_{\nu=0}^{r-1} |e^{\Omega_\nu}|^2}{\sum_{\nu=0}^{r-1} (e^{\Omega_\nu})^*}. \quad (14)$$

Therefore, the optimization problem of minimizing the error variance can be formulated as follow

$$\begin{aligned} \text{minimize: } & \sum_{\nu=0}^{r-1} \left| 1 - e^{\Omega_\nu} \frac{\sum_{k=0}^{r-1} (e^{\Omega_k})^*}{\sum_{k=0}^{r-1} |e^{\Omega_k}|^2} \right|^2 \\ \text{subject to: } & \Omega_\nu = \alpha_\nu + j(\psi_\nu - \nu\phi) \\ & (\alpha_\nu + j\psi_\nu) \in \mathcal{M} \\ & \phi \in \mathbb{R}. \end{aligned}$$

Finding the optimal Ω_ν that minimize the above expression is performed using a manipulation of closest pair problem and space partitioning algorithm [16].

VI. RESULTS

In the simulation environment, 14-bit fixed point length is considered. The investigated architectures are conventional FFT architecture and CORDIC-friendly FFT architecture. It is assumed that (90, -90, 180) degrees rotations do not require any CORDIC operations. The SQNR of each CORDIC rotations configuration is calculated according to

$$\text{SQNR} = 10 \log_{10} \left(\frac{\sum \text{Signal Power}}{\sum \text{Noise Power}} \right). \quad (15)$$

The input test signal samples are independent identically distributed complex Gaussian random variable with zero mean

and unit variance. First, the SQNR of the proposed mixed-radix FFT is compared with the conventional FFT when MSR CORDIC [4] is employed. The SQNR results of the modified and the conventional FFT schemes for different popular FFT sizes are depicted in Fig. 7. It is shown that the modified FFT scheme offers at least 30 db SQNR gain compared to the conventional FFT scheme when two-iterations MSR CORDIC is employed. Twiddle factors belonging to any FFT depend on the FFT size, as $W_N^{nk} = e^{-j2\pi nk/N}$ where N is the FFT size, $n \in \{0, 1, \dots, N-1\}$, and $k \in \{0, 1, \dots, N-1\}$. It is quite interesting to note that some twiddle factors are suitable to be synthesized via the MSR CORDIC than others. For example, twiddle factors belonging to FFT sizes whose prime factorization include (5×3) are suitable to be synthesized than the twiddle factors of the neighboring FFT sizes using MSR CORDIC. As shown in Fig. 7, FFT sizes whose prime factorization include (5×3) , such as 180, 300, 600 and 1200, achieve a superior SQNR than the neighboring FFT sizes for both conventional and modified cases.

It is worth noting that the gain offered in the SQNR is accomplished at no additional hardware cost, if SDF pipeline architecture is employed. The SQNR gain is achieved by increasing the utilization of the pipeline CORDIC in SDF architecture. When SDF FFT architectures employ the same number of CORDIC iterations in all FFT stages, large SQNR gap exists as depicted in Fig. 7 between one and two MSR iterations. Thus, different number of CORDIC iterations in each FFT stage is considered. In the investigated scenarios, the number of CORDIC iterations per FFT stage $\in \{1, 2\}$. The SQNR results for 1024-point radix-4 FFT against the total number of MSR CORDIC iterations in SDF architecture is obtained for both proposed and conventional schemes, as shown in Fig. 8. Radix-4 1024-point FFT is considered, therefore the total number of FFT stages equals 5, and hence 4 pipeline CORDICs exist in R4SDF architecture. If the total number of MSR CORDIC iterations equals 4, for example, then one MSR iteration is employed in each stage. Moreover, if the total number of MSR CORDIC iterations equals 6, then two pipeline CORDICs employ two-iteration MSR CORDIC instead, and two pipeline CORDICs employ one-iteration MSR CORDIC, respectively. The results show that for any number of total CORDIC iterations, the proposed scheme outperform the conventional scheme. Additionally, the jump in the SQNR between seven and eight total MSR iterations suggests that one stage, with less number of CORDIC iterations than other stages, is sufficient to work as bottleneck for the SQNR performance.

To assert the advantages of the proposed algorithm, various implementations of 1024-point R4SDF architecture have been designed and synthesized on IBM 130-nm CMOS technology at 100 MHz clock frequency. One design implements the conventional radix-4 FFT algorithm using MSR CORDIC. The other designs implement the proposed radix-4 CORDIC-friendly FFT algorithm which allows the use of various CORDIC types, i.e. MVR, EEAS, and MSR. Fig. 9 illustrates comparison between the SQNR performance of the investi-

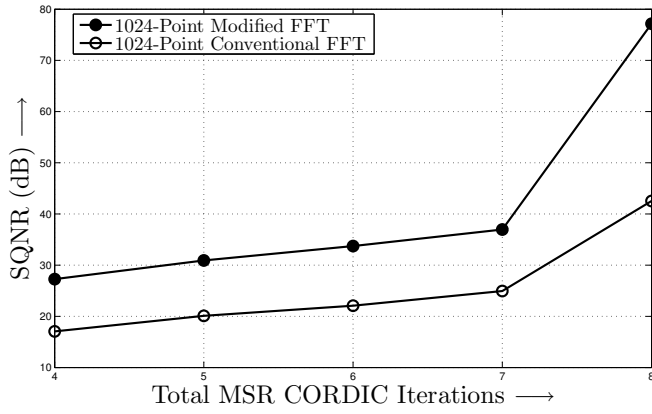


Fig. 8: SQNR achieved for modified and conventional FFT schemes against the total number of MSR CORDIC iterations in R4SDF architecture.

gated designs, where the x-axis presents the computational area, i.e. the total butterflies area as well as the CORDICs area, and y-axis presents the SQNR in dB. The proposed FFT algorithm achieves (12 – 35) dB SQNR gain compared to the conventional FFT for the same silicon area budget. Additionally, the CORDIC friendly FFT algorithm enable the use of more efficient CORDIC types such as MVR and EEAS in FFT implementation. These CORDIC types are verified to be optimal at low area budget. As shown in Fig. 9, when the SQNR required is less than 45 dB, MVR CORDIC achieves less computational area compared to the other CORDIC types. Furthermore, when the SQNR is between 45 and 63 dB, EEAS CORDIC achieves the least computational area between the investigated CORDIC types. For high SQNR application, MSR CORDIC achieves the least computational area.

VII. CONCLUSION

In this paper, a general Mixed-Radix optimization scheme of CORDIC rotations is proposed. The optimization scheme aims for choosing the optimal CORDIC-friendly rotations on the butterfly level. Instead of being bound to the original twiddle factors, the optimization scheme identify modified twiddle factors without affecting the correctness of the butterfly operation, thus offering an additional degree of freedom. The proposed optimization scheme offers (30 – 50) dB SQNR gain compared to conventional FFT scheme for different mixed-radix FFT sizes when two-iterations MSR CORDIC is employed. If SDF pipeline architecture is used, the offered SQNR gain is achieved at no additional hardware resources. Finally, the optimal CORDIC types which achieve the least silicon area are identified for different target SQNRs. MVR CORDIC is selected when the target SQNR is below 45 dB. EEAS CORDIC is selected when the SQNR required is between 45 dB and 63 dB. Finally, MSR CORDIC is chosen for high SQNR applications.

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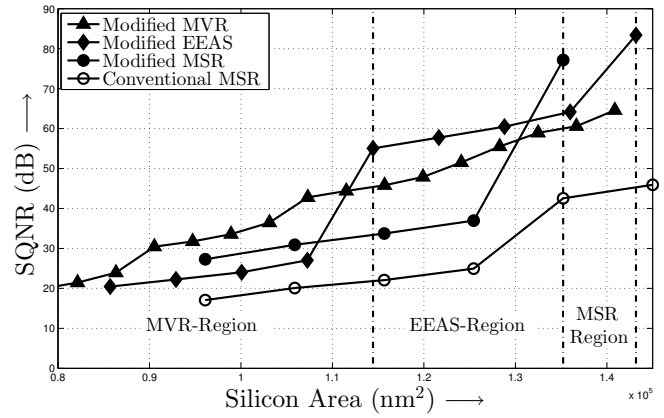


Fig. 9: SQNR performance of modified and conventional FFT schemes against the computational silicon area in 1024-point R4SDF architecture for different CORDIC types.

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